

AI Homework — Complete Notes

(AI • LLM • AI Agents • Agentic AI • Comparisons • Hierarchy)

1. Artificial Intelligence (AI)

What is AI?

AI is the branch of computer science focused on building machines/software that can perform tasks which normally require human intelligence — like understanding language, recognizing patterns, making decisions, and solving problems.

Definition of AI

AI is the simulation of human intelligence processes (learning, reasoning, self-correction) by computer systems.

Goals of AI

- Enable machines to **learn** from data/experience
- Enable machines to **reason** and make decisions
- Enable machines to **understand** language, images, sound
- Enable machines to **solve problems** and **plan**
- Automate tasks that need human-like intelligence

Characteristics of AI

- Learns from data (in most modern AI)
- Can adapt to new inputs
- Can reason/make decisions
- Can perceive environment (text, image, audio, sensors)
- Can improve performance over time (in learning systems)

Applications of AI

- Healthcare (diagnosis, drug discovery)
- Finance (fraud detection, trading)
- Education (personalized learning)
- Transportation (self-driving cars)
- Customer service (chatbots)
- Recommendation systems (Netflix, YouTube, Amazon)

Examples of AI

- Google Maps (route optimization)
- Netflix recommendation engine
- Face unlock on phones

Spam filters in email
Siri / Google Assistant

AI vs Traditional Programming

Traditional Programming	AI
Programmer writes explicit rules	System learns rules/patterns from data
Input + Program → Output	Input + Output (data) → Program (model) is learned
Fixed logic, doesn't adapt	Can adapt/improve with more data
Best for well-defined, deterministic tasks	Best for pattern recognition, ambiguity, prediction

2. LLM — Large Language Model

What is an LLM?

An LLM is a type of AI model trained on massive amounts of text data to understand and generate human-like language. It predicts the next word/token in a sequence based on patterns learned during training.

Why is it called "Large Language Model"?

Large → billions of parameters + trained on huge datasets (books, websites, articles)

Language → works primarily with human language (text)

Model → a mathematical/statistical system that maps input to output

How LLMs Work — Basic Overview

Text is broken into **tokens**

Tokens are converted into numbers (embeddings)

The model processes these using a neural network to understand context/relationships

It **predicts the next token**, one at a time, based on probability

This repeats until a full response is generated

(Note: internal architecture like Transformers is not required for this homework — just the basic flow above.)

Tokens

Small chunks of text (a word, part of a word, or punctuation)

LLMs read and generate text token by token, not full sentences at once

Example: "unhappiness" → might split into "un", "happi", "ness"

Training Data

Massive collections of text: books, websites, articles, code, conversations

The model learns grammar, facts, reasoning patterns, and language style from this data
Quality and quantity of training data directly affects the model's knowledge and behavior

Parameters

Internal numerical values (weights) the model adjusts during training

They store the "learned knowledge" of the model

More parameters generally = more capacity to learn complex patterns (not always better, but a common scaling factor)

Prompt and Response

Prompt = the input given by the user (a question, instruction, or text)

Response = the output generated by the LLM based on the prompt

LLMs generate responses by predicting the most likely next tokens given the prompt + context

Examples of LLMs

GPT (OpenAI)

Claude (Anthropic)

Gemini (Google)

LLaMA (Meta)

Applications of LLMs

Chatbots and virtual assistants

Content/text generation

Code generation and debugging help

Summarization and translation

Answering questions, tutoring/education

Limitations of LLMs

Can produce incorrect or made-up information ("hallucination")

No true understanding — works on statistical pattern prediction

Knowledge is limited to training data (has a cutoff date)

Can be biased based on training data

Struggles with precise math/logic without help

No persistent memory by default (each conversation can be independent)

3. AI Agents

What is an AI Agent?

An AI Agent is a system that can **perceive** its environment, **reason/decide** what to do, and **take action** to achieve a specific goal — often using an LLM as its "thinking" component, plus tools to interact with the world.

Perception — input/sensing module

Reasoning — the "brain" that processes information (often an LLM)

Planning — deciding steps to reach the goal

Action — executing tasks or using tools

Tools — external functions/APIs the agent can call

Memory (often included) — stores context of past steps

How an AI Agent Works (step-by-step)

Perception Agent receives input from environment — user query, sensor data, file, API response, etc.

Reasoning Agent interprets the input, understands the goal, and figures out what needs to be done.

Planning Agent breaks the goal into a sequence of steps/sub-tasks needed to complete it.

Action Agent executes the plan — this might mean calling a tool, running code, sending a request, or replying to the user.

Tools External capabilities the agent can use — e.g., web search, calculator, database, API, code execution.

Environment The space the agent operates in — could be a website, a file system, an application, or the real world (robotics).

Examples of AI Agents

A customer-support agent that reads a query, checks an order database, and replies

A coding agent that reads a bug report, edits code, and runs tests

A travel-planning agent that searches flights, compares prices, and books a ticket

A personal assistant agent that manages your calendar and sends emails

4. Agentic AI

What is Agentic AI?

Agentic AI refers to AI systems that act with a high degree of **autonomy** — they can independently set sub-goals, make decisions, use multiple tools, and carry out **multi-step tasks** with minimal human intervention, adjusting their approach based on feedback along the way.

(Simple way to put it: one agent doing one task = AI Agent. A more autonomous system that plans and executes complex, multi-step goals — possibly coordinating several agents/tools — with ongoing feedback = Agentic AI.)

Characteristics of Agentic AI

Goal-oriented behavior Works toward a defined end goal rather than just answering single queries.

Autonomy Can operate and make decisions without step-by-step human instructions.

Planning Breaks a large goal into an ordered sequence of smaller tasks.

Tool usage Actively uses external tools/APIs/software to complete tasks.

Multi-step task execution Carries out a chain of actions over time, not just a single response.

Feedback/observation loop Observes the result of each action and adjusts its next steps accordingly (plan → act → observe → re-plan).

Examples

An agentic system that researches a topic, drafts a report, checks facts, and revises it — all with minimal prompting

An autonomous coding assistant that plans a feature, writes code, tests it, fixes bugs, and repeats until done

A trip-planning system that researches options, books flights/hotels, and adjusts if a booking fails

5. Important Comparisons (likely viva questions)

AI vs ML

AI	ML
Broad field — machines simulating intelligence	Subset of AI — machines learning from data
Includes rule-based systems, ML, robotics, etc.	Specifically uses statistical algorithms to learn patterns
Goal: intelligent behavior in general	Goal: improve performance through experience/data

AI vs Generative AI

AI	Generative AI
Broad field covering all intelligent systems	Subset of AI focused on <i>creating</i> new content
Includes prediction, classification, decision-making, etc.	Generates text, images, audio, video, code
Example: spam filter (classification)	Example: ChatGPT writing an essay

AI vs LLM

AI	LLM
Umbrella field	A specific <i>type</i> of AI model
Covers vision, robotics, language, planning, etc.	Focused specifically on understanding/generating language

AI	LLM
Not all AI involves language	LLM is language-specialized AI

LLM vs AI Agent

LLM	AI Agent
A model that generates text responses	A system that perceives, decides, and <i>acts</i>
Passive — responds to a prompt	Active — can use tools, take multi-step actions
No inherent goal beyond answering	Works toward a specific goal/task
Often used <i>as the reasoning engine</i> inside an agent	Uses an LLM (often) as its "brain"

AI Agent vs Agentic AI

AI Agent	Agentic AI
A single entity performing a task	A broader system/approach involving autonomous, multi-step, adaptive behavior
May do one job (e.g., answer + one action)	Coordinates planning, tool use, feedback loops, possibly multiple agents
Lower autonomy	Higher autonomy and adaptability

Chatbot vs AI Agent

Chatbot	AI Agent
Designed mainly to converse/answer questions	Designed to <i>complete tasks</i> , not just chat
Usually no tool use or action-taking	Can use tools, take actions, interact with systems
Reactive — replies to messages	Can perceive, plan, and act toward a goal

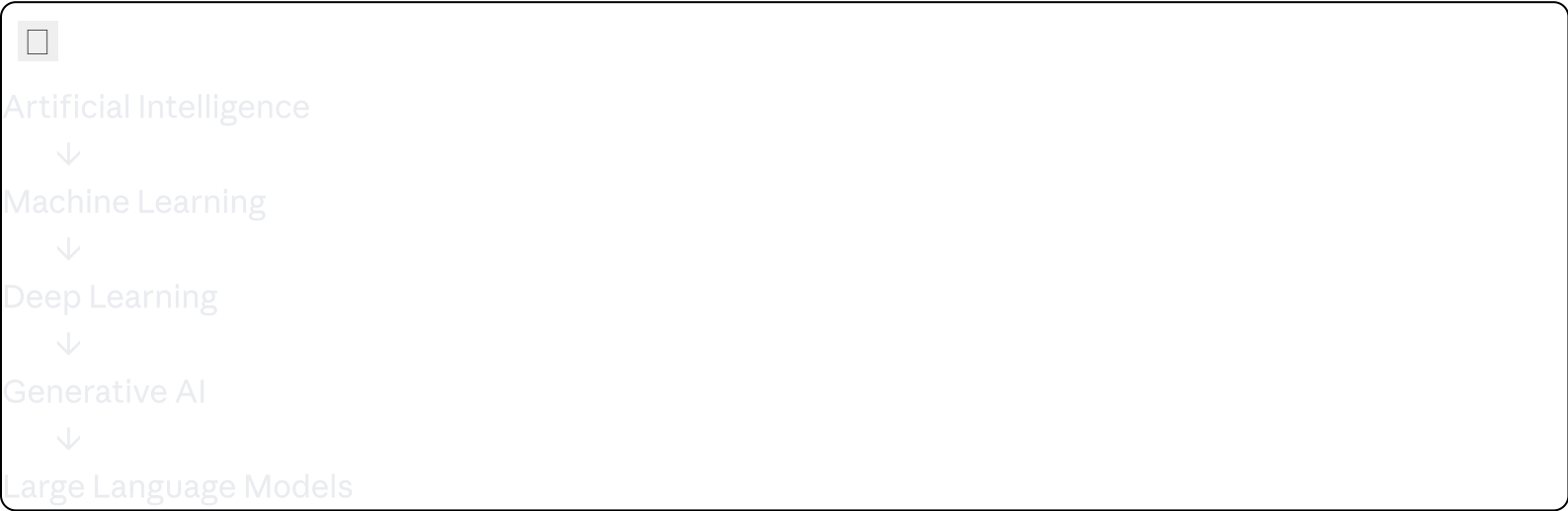
Traditional AI vs Agentic AI

Traditional AI	Agentic AI
Performs a single, narrow task (e.g., classify an image)	Performs multi-step, goal-driven tasks autonomously
No planning/tool use loop	Has planning, tool use, and feedback loops

Traditional AI	Agentic AI
Human directs each step	Operates with minimal human intervention

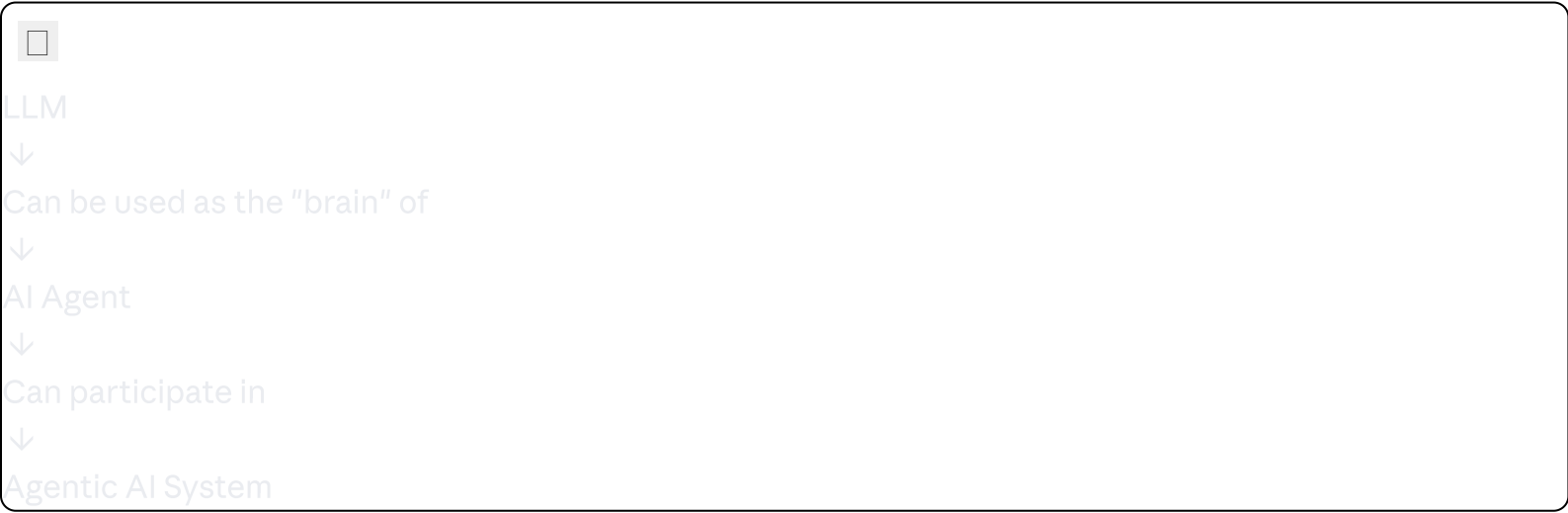
6. Basic AI Hierarchy

Field Hierarchy (broad → narrow)



- AI** — the overall field of making machines intelligent
- ML** — subset of AI where systems learn from data
- Deep Learning** — subset of ML using multi-layered neural networks
- Generative AI** — subset of Deep Learning focused on generating new content
- LLMs** — a type of Generative AI focused on language

Functional Hierarchy (how they connect in practice)



- An **LLM** provides reasoning/language ability
- An **AI Agent** wraps an LLM with perception, planning, tools, and action to do a task
- Agentic AI** is the broader autonomous system where such agents plan, act, observe, and adapt across multi-step goals — sometimes with several agents working together

Quick Viva-Prep Checklist

For each of these, be ready to: **(1) define it, (2) explain how it works in 2-3 lines, (3) give 2-3 examples, (4) compare it against the related term.**

- ☐ AI
- ☐ ML (at comparison level)
- ☐ Deep Learning (at hierarchy level)
- ☐ Generative AI (at comparison level)
- ☐ LLM
- ☐ AI Agent
- ☐ Agentic AI
- ☐ Both hierarchy diagrams (be able to draw/explain from memory)

Not required yet (per your trainer): Transformer architecture, neural-network math, RL algorithms, RAG, vector databases, LangChain.